AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT141
Introduction to Probability and Statistics

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

AOU EGYPT

Instructions:

Please read the following instructions before starting:

- Total Number of Questions: 5
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 50% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

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Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 2.5 marks, giving a total of 10 possible marks.

Q-1: A student scored 88 in an exam where the mean was 70 and the standard deviation was 6. What is the Z-score?

a) 2
b) 3
c) 4
d) 1

Q-2: Which of the following is NOT a property of the normal distribution?

a) The mean, median, and mode are equal
b) The total area under the curve equals 1
c) It is symmetric around the mean
d) It has two peaks

Q-3: A quality control team tests 200 smartphones, the probability of a faulty screen is 0.05. What is the variance?

a) 9.5
b) 10
c) 8.5
d) 12

Q-4: A group of 20 employees has a mean salary of 5,000 USD. A new employee joins with a salary of 8,200 USD. What is the new mean salary?

a) 5,400
b) 5,152.38
c) 5,200
d) 5,100

Part 2: ESSAY QUESTIONS

Each question is worth 10 marks. You should answer all FOUR questions, giving a total of 40 possible marks.

Q-1: [3+2+5 Marks]

A university registrar knows that 5% of students who register for an exam do not show up on the exam day. A faculty member randomly selects 6 registered students from the exam list. The faculty member wants to determine the probability that:

a) Exactly 3 students do not show up for the exam.
b) No students fail to show up for the exam.
c) At least 2 students do not show up for the exam.

Q-2: [3+3+4 Marks]

A continuous random variable X represents the time (in minutes) a customer spends waiting in a service queue at a government office. The variable X has a probability density function given by:

$$f(x) = k\ x^2 \qquad 0 \leq x \leq 2$$ where k is a positive constant.

Questions:

a) Calculate the value of k.
b) Find $P(X < 1)$.
c) Find $P(1 < X < 1.5)$.

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Q–3: [5+5 Marks]

a) A factory quality control engineer is monitoring the daily output (in units) of a production line, which follows a **normal distribution** with a mean of $\mu = 500$ **units** and a standard deviation of $\sigma = 20$ **units**. He wants to know what percentage of days the production line is expected to produce between **480 units** and **540 units** per day.

Hint: Use the table below

z	-1	-2.00	1.17	1.18	1	2.00	1.80	3.00
$P(Z < z)$	0.1587	0.0228	0.8790	0.8810	0.8413	0.9772	0.9641	0.9987

b) A national health research center is evaluating the effectiveness of a new medication. A sample of **25 patients** showed an average recovery time of **1,200 hours**. Assuming the population standard deviation is known to be **80 hours**, construct a **95% confidence interval** to estimate the true mean recovery time of all patients using this medication.

[Given: $Z_{0.025} = 1.96$]

Q–4: [10 Marks]

A university academic affairs office is investigating whether the average final exam score of students enrolled in an online learning program has changed from the established national baseline of **500 marks** (scaled units for consistency). A random sample of **40 students** is selected and the sample mean is found to be **495 marks**. Given that the population standard deviation is **10 marks**, test the hypothesis at the **0.05 significance level** to determine if there is evidence of a difference in the average exam score.

[Given: $Z_{0.025} = 1.96$]

End of questions

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